INAYAT RAHIM

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EDUCATION

SZABIST University Islamabad Islamabad
Bachelor of Science (B.S) Artificial Intelligence *2023 – 2027*

EXPERIENCE

ML Intern (Computer Vision & Bio-AI) Remote
ARCH Technologies *April 2024 – Aug 2024*

- Engineered a lightweight transformer-based segmentation pipeline with Segment Anything (SAM) to annotate and analyze high-resolution satellite imagery for an environmental AI system.
- Co-developed biomedical GNN models to predict gene-disease links using multimodal omics data and large-scale biomedical knowledge graphs.
- Built modular preprocessing tools and visualizers for protein interaction prediction using RDKit, Py3Dmol, and AlphaFold outputs.
- Managed experiment tracking and model deployment using Hugging Face Hub and Weights & Biases across both geospatial and molecular AI pipelines.

COMPUTER VISION PROJECTS

AI-Powered Perimeter Surveillance using Multispectral Vision | *Swin Transformer, GroundingDINO, MobileSAM, OpenCV Thermal SDK*

- Description: Developed an edge-based surveillance system that fuses thermal and RGB data to detect intrusions, vehicles, and weapon outlines under poor visibility.
- Role: Designed a vision transformer pipeline with GroundingDINO + Swin Transformer for object detection in night and fog scenarios.

Vision Transformers for Multimodal Manufacturing Defect Detection | *ViT, SAM, Swin Transformer, Gradio, Label Studio, PaddlePaddle*

- Description: Developed ViT-based pipeline to detect surface anomalies in manufacturing units using visible + thermal imaging.
- Role: Combined SAM with Swin Transformer backbones for precise segmentation in industrial environments.

AI-Powered Geo-Intelligence Platform for Defense & Aerospace | *SAM, PaddleRS, GeoTransformer, Multispectral Imaging, FastSAM*Dec

- Description: Designed a dual-view vision system to analyze multispectral satellite data for terrain, object detection, and mission-critical decision-making.
- Role: Integrated Meta's Segment Anything with transformer-based encoders to track real-time spatial changes across AOIs.

CERTIFICATIONS

Advanced Computer Vision & Transformer Models – Hugging Face & Meta AI Feb 2025

- Specialized in multimodal transformers (ViT, SAM, Owl-ViT, CLIP) for advanced vision tasks.
- Explored LoRA, quantization, AdapterFusion, and zero-shot vision with real-world geospatial + industrial datasets.
- Tech Used: SAM, CLIP, ViT, Gradio, Hugging Face Transformers, PyTorch

Geospatial Vision AI – Satellite Vision Lab + PaddlePaddle Open Source June 2025

- Trained vision transformer models on hyperspectral and multispectral satellite data for wildfire prediction, deforestation mapping, and climate analytics.
- Worked with time-series spatial modeling using Earthformer and segmentation with SAM & PaddleRS.
- Tech Used: PaddleRS, Sentinel Hub, Earthformer, SAM, GeoContrast

AI for Scientific Imaging & Industrial Monitoring – DeepMind-inspired Curriculum May 2025

- Built AI pipelines using AlphaFold, BioNeMo, and OpenFold for protein structure + manufacturing materials analysis.
- Learned scalable AI deployment across industrial, biomedical, and aerospace imaging.
- Tech Used: AlphaFold, BioNeMo, PaddleHelix, Py3Dmol, Hugging Face

Technical Skills

Computer Vision: Segment Anything (SAM), GroundingDINO, FastSAM, GeoContrast, Swin Transformer, ViT, CLIP, PaddleRS, Earthformer, Gradio
Frameworks/Libraries: PaddlePaddle, PyTorch, Hugging Face Transformers, DGL (Deep Graph Library), RDKit, Py3Dmol
Tools/APIs: Sentinel Hub API, Streamlit, Label Studio, Hugging Face Spaces, Git
Deployment: Gradio, Hugging Face Spaces, Streamlit, Docker (basics)
Domains: Satellite Imagery, Spatiotemporal AI, Geospatial Vision, Biomedical CV, Transformer-based Imaging

Academic Achievements

- Microsoft Learn Student Ambassador (MLSA) – Beta Tier
- Selected Participant – Cohere Lab Summer School on NLP, Cutting Edge Multi Disciplinary Transformers Advanced Generative AI
- Awarded $1,000 in Azure Credits – Microsoft for Startups Founders Hub
- Received Cloud AI Credits – OpenAI, GitHub Copilot, Notion, Miro, and more
- Aspire Leaders Program – Led by Harvard Business School (2025 Cohort)

Volunteer Experience

- Organized and led Python and AI workshops at university and community level.
- Mentored students in AI project building, research, and tech career pathways.
- Contributed to open-source AI models, tools, and learning platforms; developed educational AI notebooks and supported community-driven projects across multimodal, geospatial, biomedical, and computer vision domains.

References

Taimur Shahzad Islamabad, Pakistan

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